

CARMA-Based Deseasonalization, Calibration and Pricing in Renewable Energy Markets

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Abstract

1 Introduction

Electricity prices are increasingly exposed to weather and renewable generation. Temperature affects demand, while wind and solar production affect both the level and the intraday shape of spot prices. Energy contracts written on such markets therefore carry hybrid risk, since their value depends on the joint behavior of electricity prices and physical factors that are non-traded or only partially traded.

This paper develops an empirical CARMA framework for these joint risks. We first remove deterministic seasonal patterns from German day-ahead prices, temperature, wind capacity factors, and solar capacity factors. The resulting stochastic factors are then modeled with continuous-time autoregressive moving-average dynamics, which provide a state-space representation for calibration, dependence estimation, and pricing.

The preprocessing is factor-specific because the deterministic component differs across markets and weather variables. Electricity prices contain calendar and intraday effects, temperature has smooth daily and annual cycles, and wind capacity factors are naturally treated through a logit transformation. Solar requires an additional clear-sky normalization before the latent logit transformation, as night hours and moving sunrise and sunset times induce deterministic boundary effects.

The contribution is threefold. First, we build a consistent preprocessing pipeline for electricity, temperature, wind, and solar data. Second, we calibrate marginal CARMA models and recover their driving noise, allowing heavy-tailed Levy specifications when Gaussian innovations are insufficient. Third, we use the calibrated factors to estimate price-weather and price-renewable dependence channels and to price energy contracts exposed to these joint risks.

The paper is organized as follows. Section 2 describes the data transformations and the construction of the stochastic factors. Section 3 presents the marginal CARMA calibration and driver diagnostics. Section 4 estimates the dependence channels. Section 5 discusses the pricing implementation, and Section 6 concludes.

2 Data and Deseasonalization

2.1 Electricity Prices

Data. We use hourly German day-ahead electricity prices for the DE-LU bidding zone. The calibration sample runs from 1 January 2023 to 31 December 2025 and is indexed in UTC.

The 2021–2025 history is shown in Figure 1. The 2021–2022 period contains exceptional crisis spikes, while the 2023–2025 window keeps a recent regime with both positive and negative price spikes. Since the German spot market can contain negative prices, the price is first shifted before applying the logarithm:

$$\tilde{S}_t = \log(S_t + \delta), \quad \delta = 1000 \text{ EUR/MWh}. \quad (1)$$

The shift is only used to define a positive log-price factor. Physical prices are recovered by the inverse map $S_t = \exp(\tilde{S}_t) - \delta$.

Deseasonalization. The price model is specified in the shifted log-price coordinate:

$$\tilde{S}_t = \Lambda_t^P + X_t^P. \quad (2)$$

Here Λ_t^P is deterministic and X_t^P is the stochastic price factor later modeled by CARMA dynamics. We estimate Λ_t^P with a Paraschiv-style seasonal weight in the transformed coordinate. Let $\bar{p}$ be the sample mean of $\tilde{S}_t$, and let d be the calendar day containing hour t . The fitted seasonal component is

$$\hat{\Lambda}_t^P = \bar{p} \hat{F}_d^Y \hat{F}_t^D, \quad \hat{X}_t^P = \tilde{S}_t - \hat{\Lambda}_t^P. \quad (3)$$

The day-level factor $\hat{F}_d^Y$ is estimated from daily average log-prices. If $\bar{S}_d^{\log}$ is the daily mean of $\tilde{S}_t$, the observed factor is

$$F_d^{Y,\text{obs}} = \frac{\bar{S}_d^{\log}}{\text{yearly mean of } \bar{S}_d^{\log}}. \quad (4)$$

We regress $F_d^{Y,\text{obs}}$ on weekday and month indicators, with August split into early and late subperiods. The fitted daily values are broadcast to the hourly grid.

The intraday factor is estimated on 20 profile classes. Weekdays are grouped by month, while Saturdays and Sundays are grouped by season. Within a profile class c , the shifted log-price is regressed on hour-of-day indicators:

$$\tilde{S}_t = a_0^{(c)} + \sum_{h=1}^{23} a_h^{(c)} \mathbf{1}_{\{\text{hour}(t)=h\}} + \varepsilon_t^{(c)}, \quad t \in c. \quad (5)$$

The fitted intraday curve is normalized by its class mean, which gives $\hat{F}_t^D$. This normalization makes $\hat{F}_t^D$ a shape factor rather than a level factor.

This construction removes the main calendar, monthly, and hourly price patterns before the CARMA calibration. The remaining series $\hat{X}^P$ is treated as the deseasonalized stochastic log-price factor.

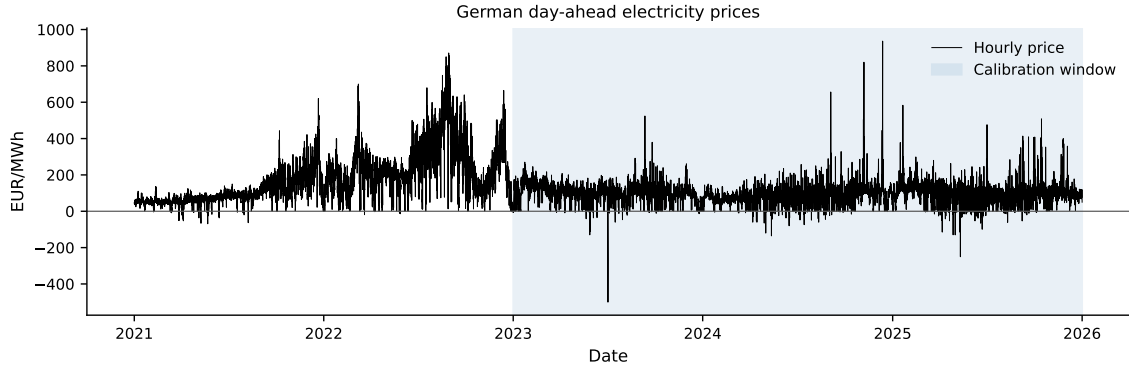


Figure 1: Hourly German day-ahead electricity prices from 2021 to 2025. The shaded region is the 2023–2025 calibration window used for the price deseasonalization and CARMA calibration.

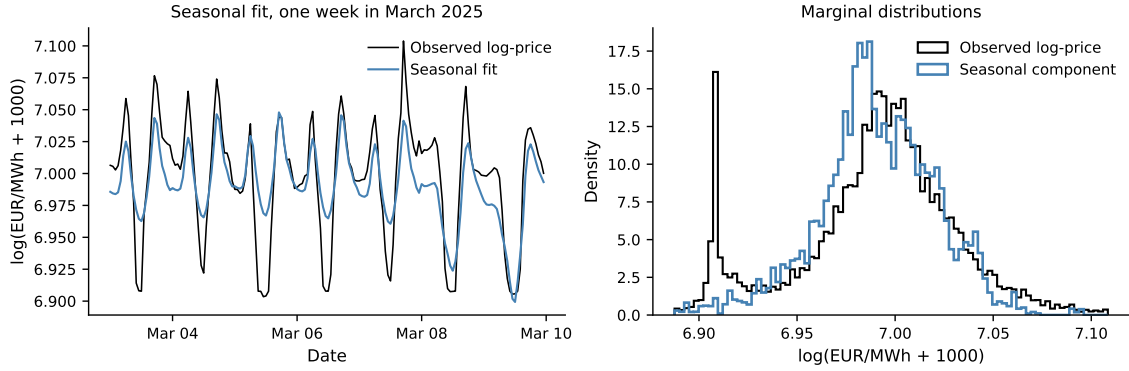


Figure 2: Price deseasonalization diagnostics. Left: observed shifted log-price and fitted deterministic component over one week in March 2025. Right: marginal distributions of the shifted log-price and of the fitted deterministic component over the calibration sample.

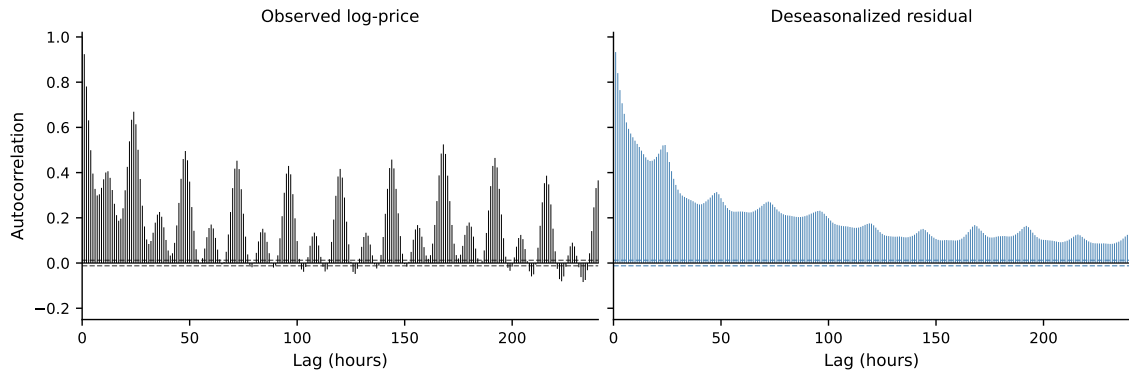


Figure 3: Empirical autocorrelation of the shifted log-price and of the deseasonalized price residual over 240 hourly lags. The deterministic deseasonalization removes the main calendar peaks, while the remaining serial dependence motivates the CARMA calibration.

2.2 Temperature

Data. We use hourly 2-meter temperature for Berlin over the same 2023–2025 UTC sample. The temperature factor is modeled in physical units, without a positivity transform:

$$T_t = \Lambda_t^T + X_t^T, \quad (6)$$

where Λ_t^T is deterministic seasonality and X_t^T is the deseasonalized stochastic temperature factor.

Deseasonalization. The deterministic component is estimated by a Fourier regression with a linear trend, a daily cycle, an annual cycle, and interactions between the two cycles. Let τ_t be the number of hours elapsed since the beginning of the sample and set $P_y = 365.25 \times 24$. The seasonal component is

$$\begin{aligned} \Lambda_t^T(\theta) = & \beta_0 + \beta_1 \tau_t + \sum_{k=1}^3 \left[a_k \cos\left(\frac{2\pi k \tau_t}{24}\right) + b_k \sin\left(\frac{2\pi k \tau_t}{24}\right) \right] \\ & + a_y \cos\left(\frac{2\pi \tau_t}{P_y}\right) + b_y \sin\left(\frac{2\pi \tau_t}{P_y}\right) \\ & + \sum_{k=1}^3 \left[u_k \cos\left(\frac{2\pi k \tau_t}{24}\right) + v_k \sin\left(\frac{2\pi k \tau_t}{24}\right) \right] \cos\left(\frac{2\pi \tau_t}{P_y}\right) \\ & + \sum_{k=1}^3 \left[\tilde{u}_k \cos\left(\frac{2\pi k \tau_t}{24}\right) + \tilde{v}_k \sin\left(\frac{2\pi k \tau_t}{24}\right) \right] \sin\left(\frac{2\pi \tau_t}{P_y}\right). \end{aligned} \quad (7)$$

The first sum captures the deterministic intraday pattern, the annual sine and cosine capture the low-frequency seasonal level, and the last two sums allow the intraday profile to vary smoothly over the year. All coefficients are estimated by ordinary least squares:

$$\hat{\theta} = \arg \min_{\theta} \sum_t (T_t - \Lambda_t^T(\theta))^2, \quad \hat{X}_t^T = T_t - \Lambda_t^T(\hat{\theta}). \quad (8)$$

Figure 4 shows the fitted deterministic component over the same week used for the price diagnostics and compares the marginal distributions of observed and seasonal temperature. The ACF comparison in Figure 5 shows that the deterministic regression removes the dominant periodic component, while the residual retains serial dependence to be modeled by the temperature CARMA process.

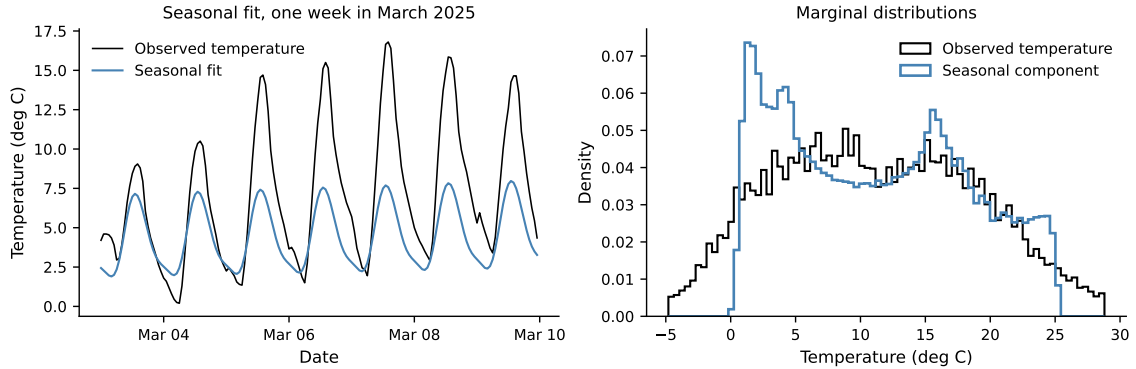


Figure 4: Temperature deseasonalization diagnostics. Left: observed temperature and fitted deterministic component over one week in March 2025. Right: marginal distributions of observed temperature and of the fitted deterministic component over the calibration sample.

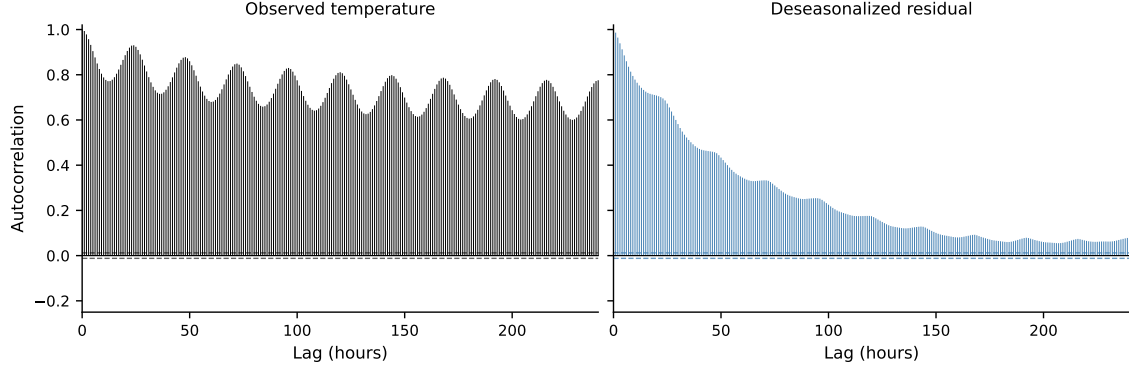


Figure 5: Empirical autocorrelation of observed temperature and of the deseasonalized temperature residual over 240 hourly lags.

2.3 Solar Capacity Factors

Data and clear-sky proxy. We use hourly German solar capacity factors $Q_t \in [0, 1]$. Solar production has structural zeros at night and a daylight window that changes over the year, as shown in Figure 6. These features make a direct additive regression on Q_t inappropriate, since the seasonal level and the support of the process vary jointly over the calendar.

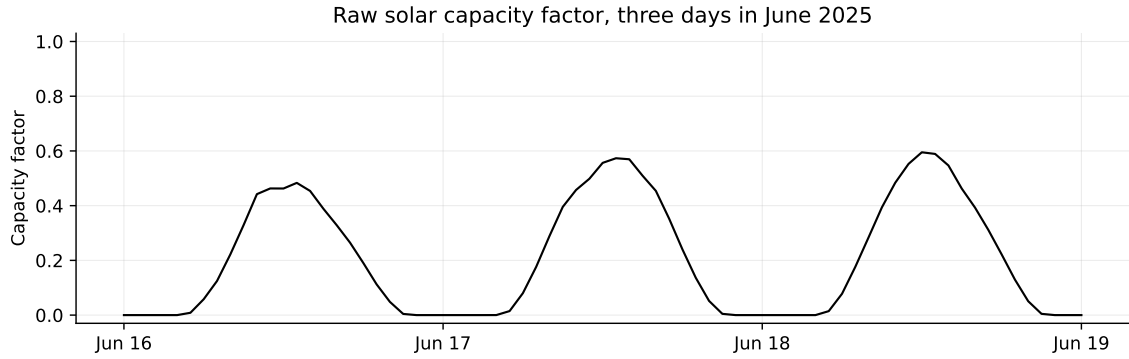


Figure 6: Raw hourly German solar capacity factor over three days in June 2025. The series combines deterministic daylight availability, cloud-driven reductions during daylight hours, and structural nighttime zeros.

We therefore first estimate an empirical clear-sky proxy. Let $d(t)$ denote the day of year and $h(t)$ the hour of day. For each calendar cell (d, h) , the raw clear-sky level is defined as the upper-tail climatology

$$C_{d,h}^{\text{raw}} = q_{0.98}\{Q_t : d(t) = d, h(t) = h\}, \quad (9)$$

where $q_{0.98}$ is the empirical 98% quantile. Missing cells are filled by coarser calendar averages, and each hourly day-of-year profile is smoothed with a circular 21-day moving average. The resulting proxy is denoted

$$\hat{C}_t = \hat{C}_{d(t),h(t)}. \quad (10)$$

This proxy is an empirical capacity-factor envelope. It captures the seasonal and intraday availability of solar radiation while keeping the subsequent stochastic factor bounded.

Latent transformation and deseasonalization. The stochastic solar factor is defined from the shortfall of production relative to the clear-sky proxy:

$$R_t = \text{clip}_{[0,1]} \left(1 - \frac{Q_t}{\max(\hat{C}_t, \varepsilon_C)} \right), \quad \varepsilon_C = 10^{-4}. \quad (11)$$

The bounded risk driver is then mapped to a real-valued latent coordinate by a logit transform,

$$Z_t = \text{clip}_{[\varepsilon, 1-\varepsilon]} \left(\frac{R_t - \alpha}{\beta} \right), \quad Y_t^Q = \log \left(\frac{Z_t}{1 - Z_t} \right), \quad (12)$$

with $\alpha = 10^{-4}$, $\beta = 0.9998$, and $\varepsilon = 10^{-4}$ in the present calibration. Deseasonalization is performed in this latent coordinate:

$$Y_t^Q = \Lambda_t^Q + X_t^Q. \quad (13)$$

The deterministic latent component is estimated by ordinary least squares. It combines hour indicators, month indicators, annual Fourier terms, and hour-by-annual interactions. With $H_h(t) = \mathbf{1}_{\{h(t)=h\}}$, $M_m(t) = \mathbf{1}_{\{\text{month}(t)=m\}}$, and $\phi_t = 2\pi d(t)/365.25$, we use

$$\begin{aligned} \Lambda_t^Q(\theta) = & \gamma_0 + \sum_{h=1}^{23} \gamma_h H_h(t) + \sum_{m=2}^{12} \eta_m M_m(t) \\ & + \sum_{k=1}^3 [a_k \cos(k\phi_t) + b_k \sin(k\phi_t)] \\ & + \sum_{h=1}^{23} H_h(t) \sum_{k=1}^2 [u_{h,k} \cos(k\phi_t) + v_{h,k} \sin(k\phi_t)]. \end{aligned} \quad (14)$$

The residual is

$$\hat{\theta} = \arg \min_{\theta} \sum_t (Y_t^Q - \Lambda_t^Q(\theta))^2, \quad \hat{X}_t^Q = Y_t^Q - \Lambda_t^Q(\hat{\theta}). \quad (15)$$

For visualization in physical units, the fitted latent seasonality is mapped back to capacity factors through

$$\hat{Q}_t^{\text{det}} = \hat{C}_t \left[1 - \alpha - \beta \sigma(\Lambda_t^Q(\hat{\theta})) \right], \quad \sigma(x) = \frac{1}{1 + \exp(-x)}. \quad (16)$$

Figure 7 displays the main transformations over one week. The top panel shows observed production, the clear-sky envelope, and the latent seasonal component mapped back to capacity-factor scale. The middle panel shows the bounded shortfall and its normalized version. The bottom panel shows the latent coordinate Y_t^Q and its fitted seasonality.

Figure 8 gives the corresponding one-week fit and compares the marginal distributions of the latent coordinate and fitted latent deterministic component. The ACF comparison in Figure 9 shows the reduction in calendar-driven dependence after the latent deseasonalization.

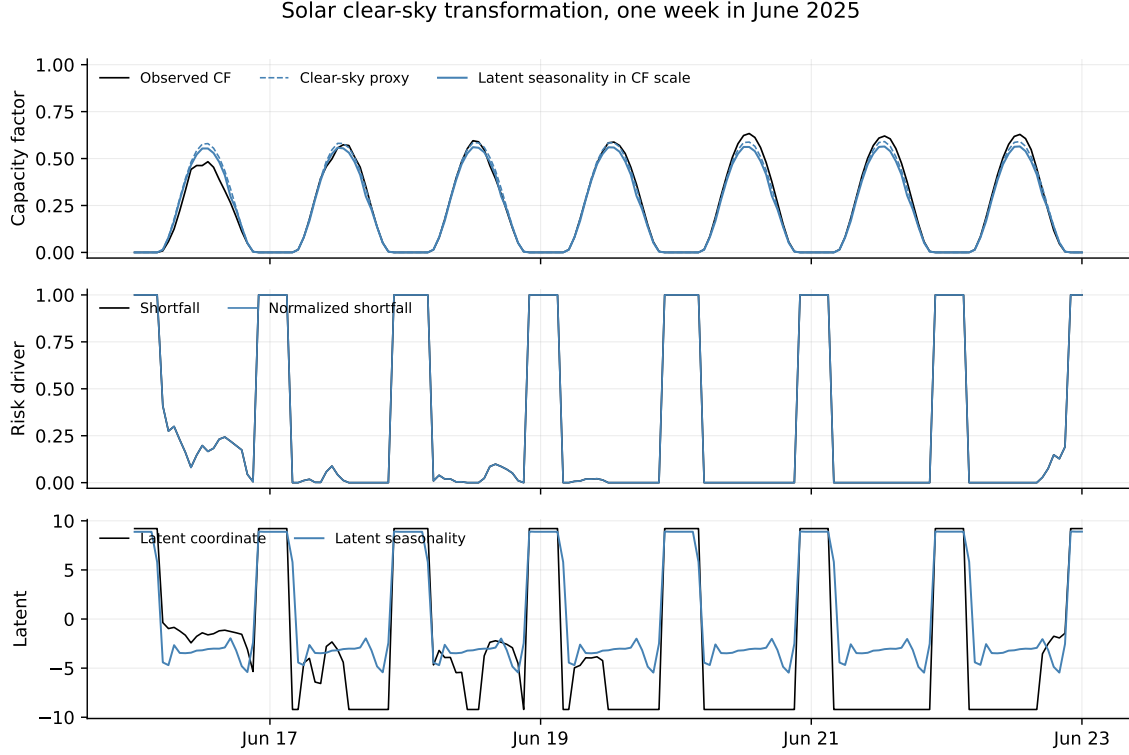


Figure 7: Solar clear-sky transformation over one week in June 2025. Top: observed capacity factor, clear-sky proxy, and latent seasonality mapped to capacity-factor scale. Middle: shortfall and normalized shortfall. Bottom: latent coordinate and latent seasonality.

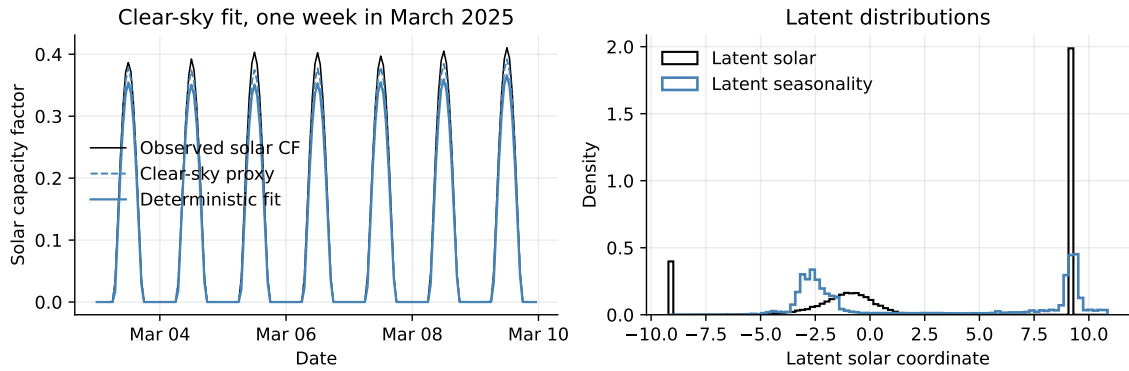


Figure 8: Solar clear-sky deseasonalization diagnostics. Left: observed solar capacity factor, clear-sky proxy, and latent seasonality mapped to capacity-factor scale over one week in March 2025. Right: marginal distributions of the latent coordinate and latent seasonality.

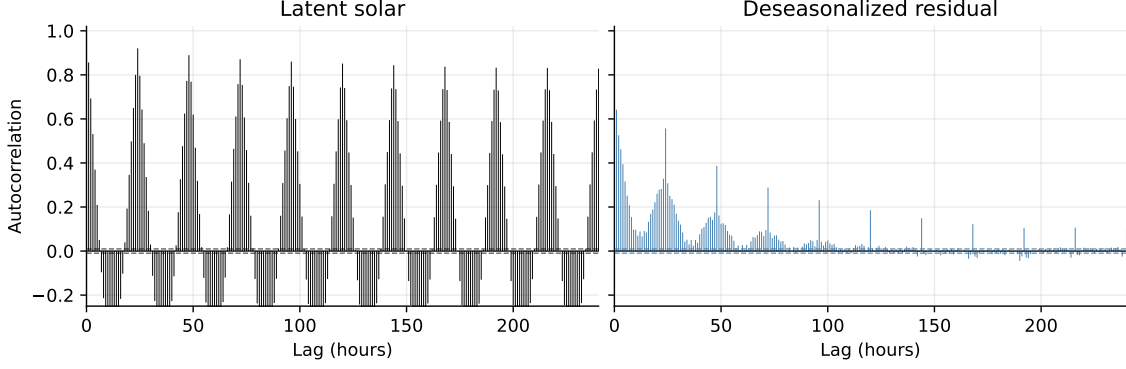


Figure 9: Empirical autocorrelation of the latent solar coordinate and of the deseasonalized solar residual over 240 hourly lags.

2.4 Wind Capacity Factors

Data and bounded transform. We use hourly German wind capacity factors $W_t \in [0, 1]$, aligned on the same UTC calendar as the German price panel. Wind does not have structural nighttime zeros, but the capacity factor is bounded and can spend long periods near low production levels. Figure 10 shows the raw series over three days. The model is therefore specified after a logit transformation of the clipped capacity factor:

$$W_t^\varepsilon = \text{clip}_{[\varepsilon, 1-\varepsilon]}(W_t), \quad Y_t^W = \log\left(\frac{W_t^\varepsilon}{1 - W_t^\varepsilon}\right), \quad \varepsilon = 10^{-4}. \quad (17)$$

The clipping only avoids infinite logit values at the boundaries. The inverse map used for physical diagnostics is

$$W_t = \sigma(Y_t^W), \quad \sigma(x) = \frac{1}{1 + \exp(-x)}. \quad (18)$$

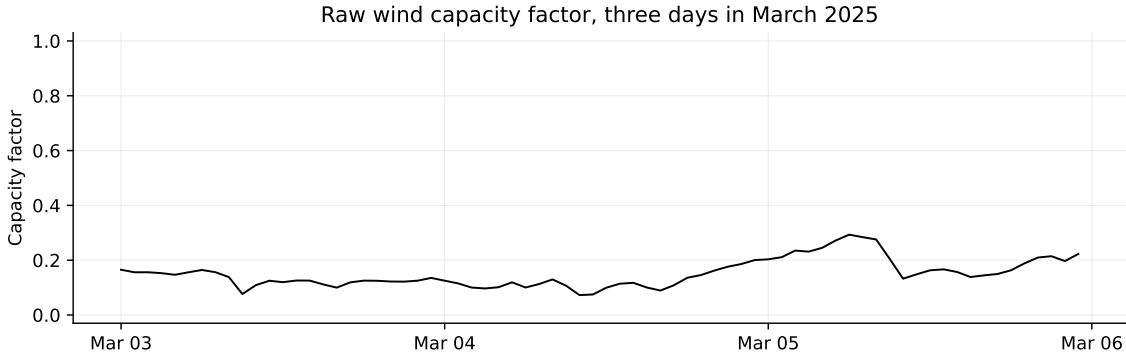


Figure 10: Raw hourly German wind capacity factor over three days in March 2025. The process is bounded, persistent, and irregular, without the deterministic day-night support constraint present in solar production.

Deseasonalization. Deseasonalization is performed in the logit coordinate:

$$Y_t^W = \Lambda_t^W + X_t^W. \quad (19)$$

The deterministic component combines intraday, weekly, monthly, and annual calendar effects. Let $H_h(t) = \mathbf{1}_{\{\text{hour}(t)=h\}}$, $D_j(t) = \mathbf{1}_{\{\text{weekday}(t)=j\}}$, $M_m(t) = \mathbf{1}_{\{\text{month}(t)=m\}}$, and $\phi_t =$

$2\pi d(t)/365$, where $d(t)$ is the day of year. The fitted seasonality is

$$\begin{aligned} \Lambda_t^W(\theta) = & \gamma_0 + \sum_{h=1}^{23} \gamma_h H_h(t) + \sum_{j=1}^6 \delta_j D_j(t) + \sum_{m=2}^{12} \eta_m M_m(t) \\ & + \sum_{k=1}^2 [a_k \sin(k\phi_t) + b_k \cos(k\phi_t)]. \end{aligned} \quad (20)$$

All coefficients are estimated by ordinary least squares:

$$\hat{\theta} = \arg \min_{\theta} \sum_t (Y_t^W - \Lambda_t^W(\theta))^2, \quad \hat{X}_t^W = Y_t^W - \Lambda_t^W(\hat{\theta}). \quad (21)$$

The blue physical-scale curve in the diagnostics is not fitted directly on W_t . It is the logit seasonality mapped back as $\hat{W}_t^{\text{sea}} = \sigma(\Lambda_t^W(\hat{\theta}))$.

Figure 11 gives the one-week physical-scale view and compares the marginal distributions in the logit coordinate. The ACF comparison in Figure 12 shows that calendar deseasonalization removes part of the deterministic periodicity, while the remaining serial dependence is left for the wind CARMA calibration.

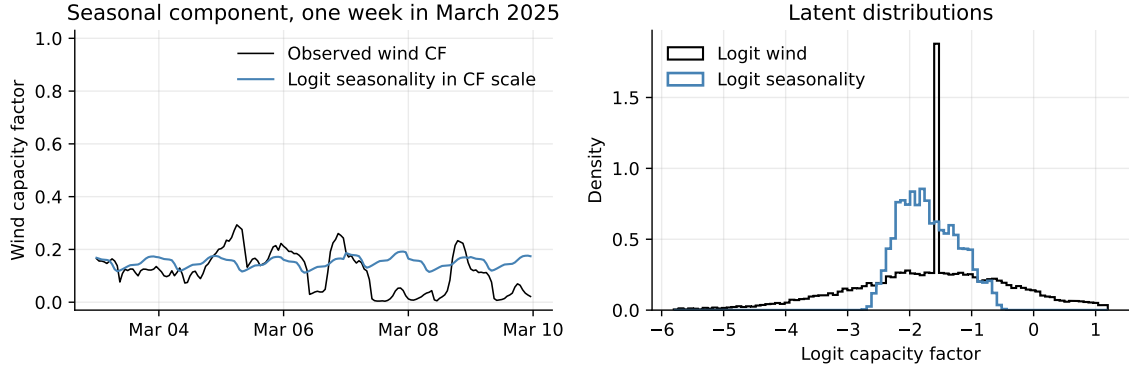


Figure 11: Wind deseasonalization diagnostics. Left: observed wind capacity factor and fitted logit seasonality mapped to capacity-factor scale over one week in March 2025. Right: marginal distributions of the logit capacity factor and fitted logit seasonality.

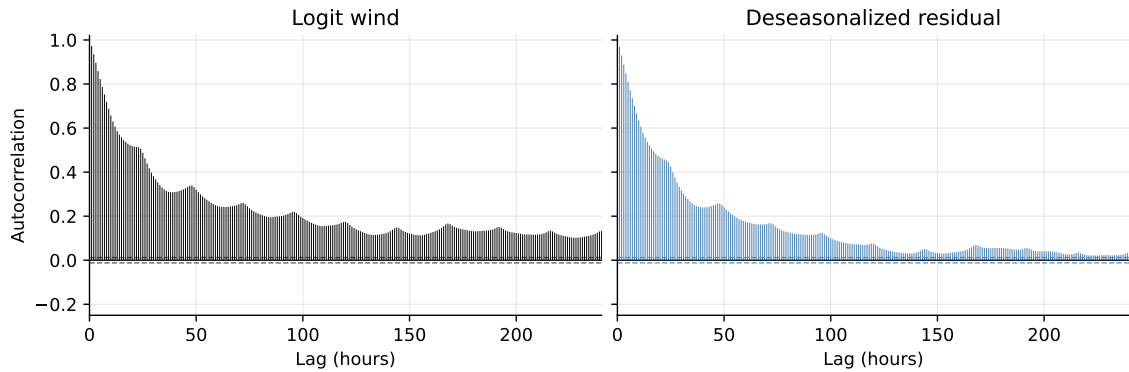


Figure 12: Empirical autocorrelation of the logit wind capacity factor and of the deseasonalized wind residual over 240 hourly lags.

3 Marginal CARMA Calibration

4 Coupling Calibration

5 Pricing

6 Conclusion

References